EPS Rev B — integrated development schematic

Checkpoint: controller, CAN, essential 3V3 and recovery converter. Source/interlock, BMS, charger and protection pending.

Controller

File: controller.kicad_sch

CAN interfaces

File: can_interfaces.kicad_sch

Essential 3V3

File: essential_3v3.kicad_sch

Recovery 5V converter

File: recovery_5v.kicad_sch

Reference ranges: 100 controller, 200 BMS, 300 charger, 400 essential 3V3, 500 recovery 5V, 600 protection.

External interface names preserved. Logical connectors are placeholders; no production connector acceptance.

Incomplete ERC remains visible. No PCB synchronization or fabrication acceptance.

S03 severity API unavailable; S11 required field display unsupported; reused custom symbol S08 review pending.

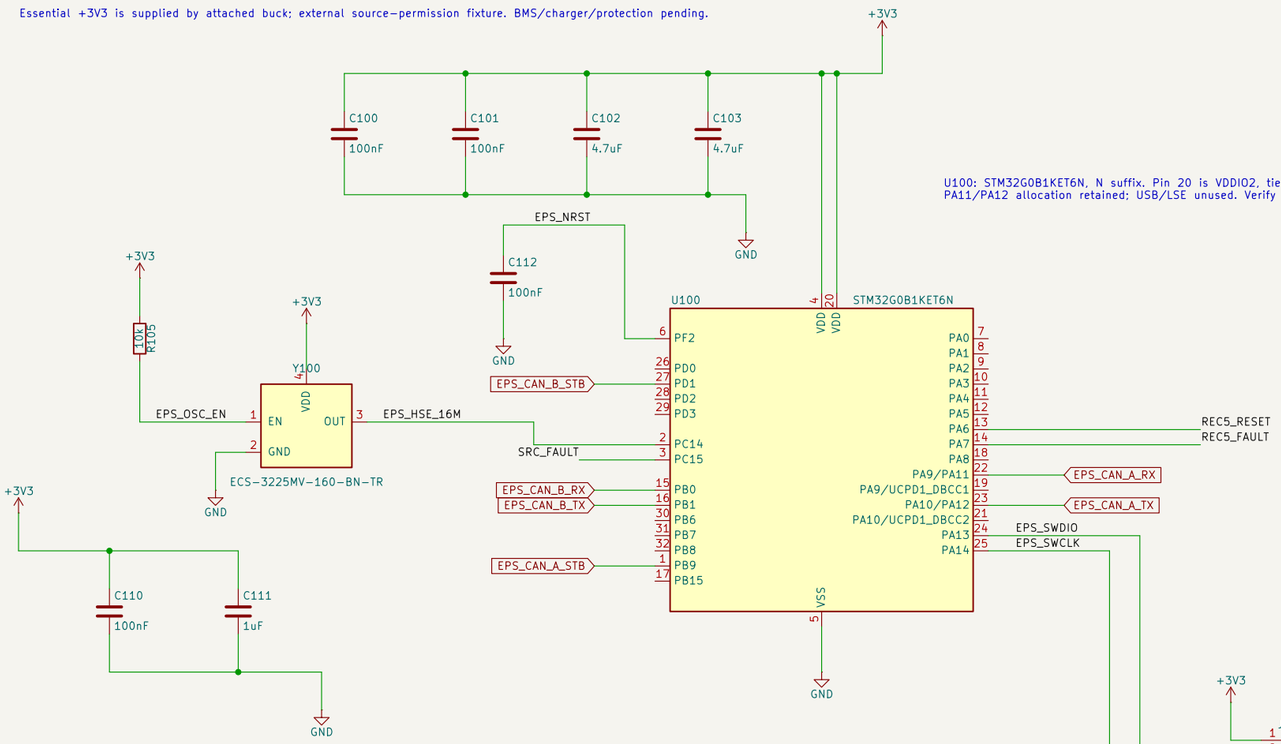
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CONTROLLER — integrated development checkpoint

Essential +3V3 is supplied by attached buck; external source-permission fixture. BMS/charger/protection pending.

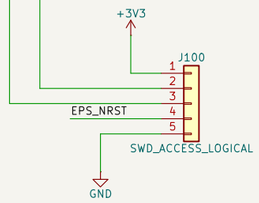


16 MHz HSE bypass; no passive crystal.
Place C110/C111 locally at Y100 in layout; source-series option open.

U100: STM32G0B1KET6N, N suffix. Pin 20 is VDDIO2, tied to VDD.
PA11/PA12 allocation retained; USB/LSE unused. Verify UCPD reset configuration.

PROTECTION INTERFACES — NOT YET LINKED
PA6 REC5_RESET -> existing 47k Q107 base path in protection study.
PA7 REC5_FAULT; PC15 SRC_FAULT: active-low, 47k essential pullups there.
Reset: 2 ms pulse, >=10 s cooldown and rail qualification.
Reset is not permission to bypass protection.

J100: logical SWD access; physical connector unselected.
1 VTref sense only, 2 SWDIO, 3 SWCLK, 4 NRST, 5 GND.
No injected probe power. Power-off/backfeed review open.



OPEN: ADC conditioning, local charger I2C, isolated pack SMBus,
UART, interlock status and payload request. Required pins remain unconnected.
No EPS drive on IHU-owned COMMS_EN / PAYLOAD_EN or stack alert.

PD3 reserved. Do not connect BMS BTP directly to 3V3 logic.
Unused BTP is tied to BMS VSS in the proposed BMS circuit; poll pack status.

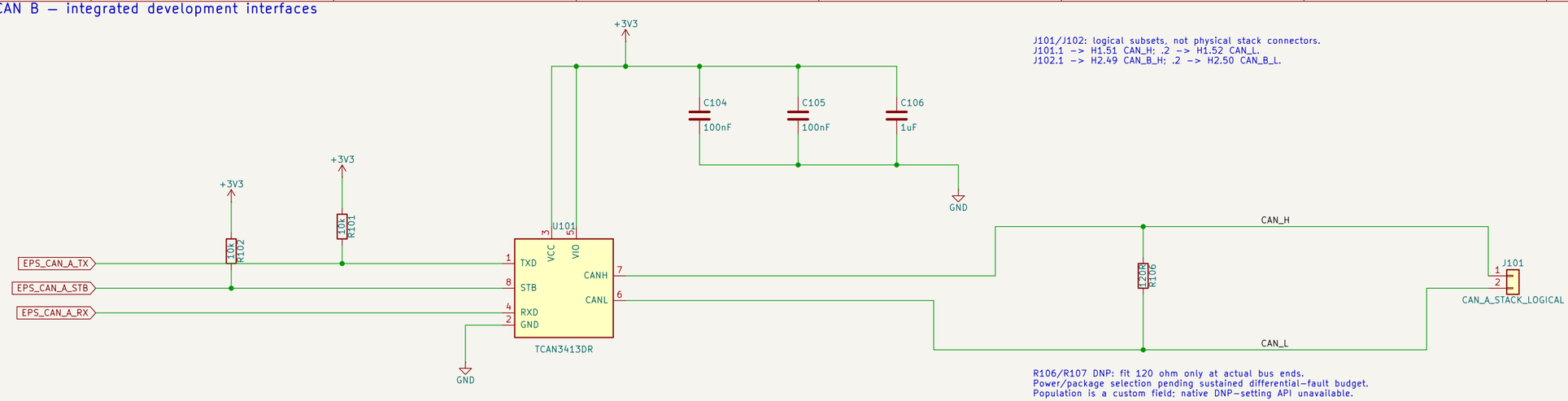
S11 OPEN: metadata is saved; required visible field styling unsupported.
S08: reused TCAN/ECS custom symbols remain under original construction review.

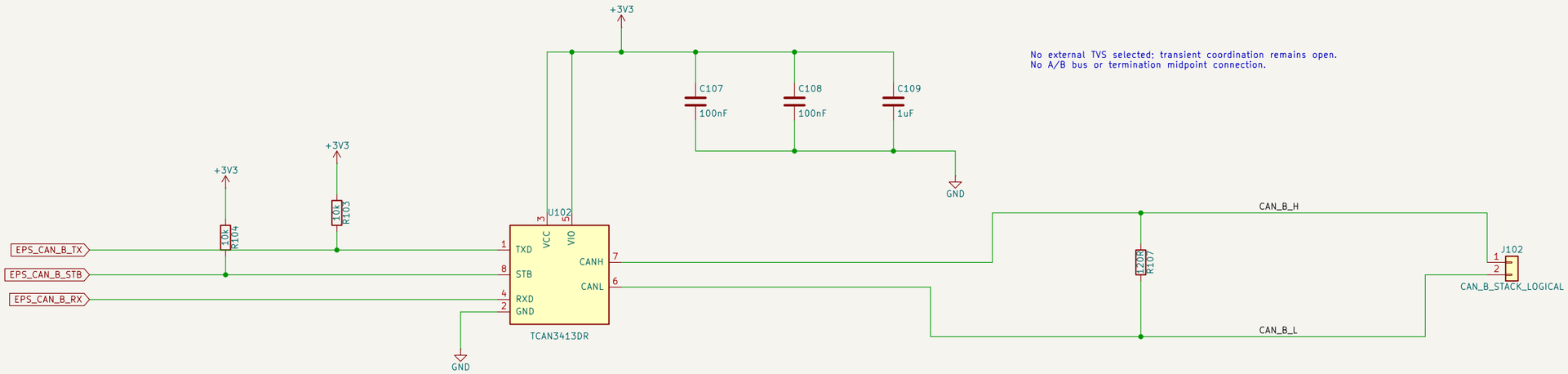
Sheet: /Controller/
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CAN A / CAN B — integrated development interfaces





FAULT LIMIT: shared +3V3, GND, MCU and clock remain common paths.
Independent VCC/VIO + pullup branch limiting remains a written candidate.
CAN B is bus-path recovery, not independent controller/power redundancy.

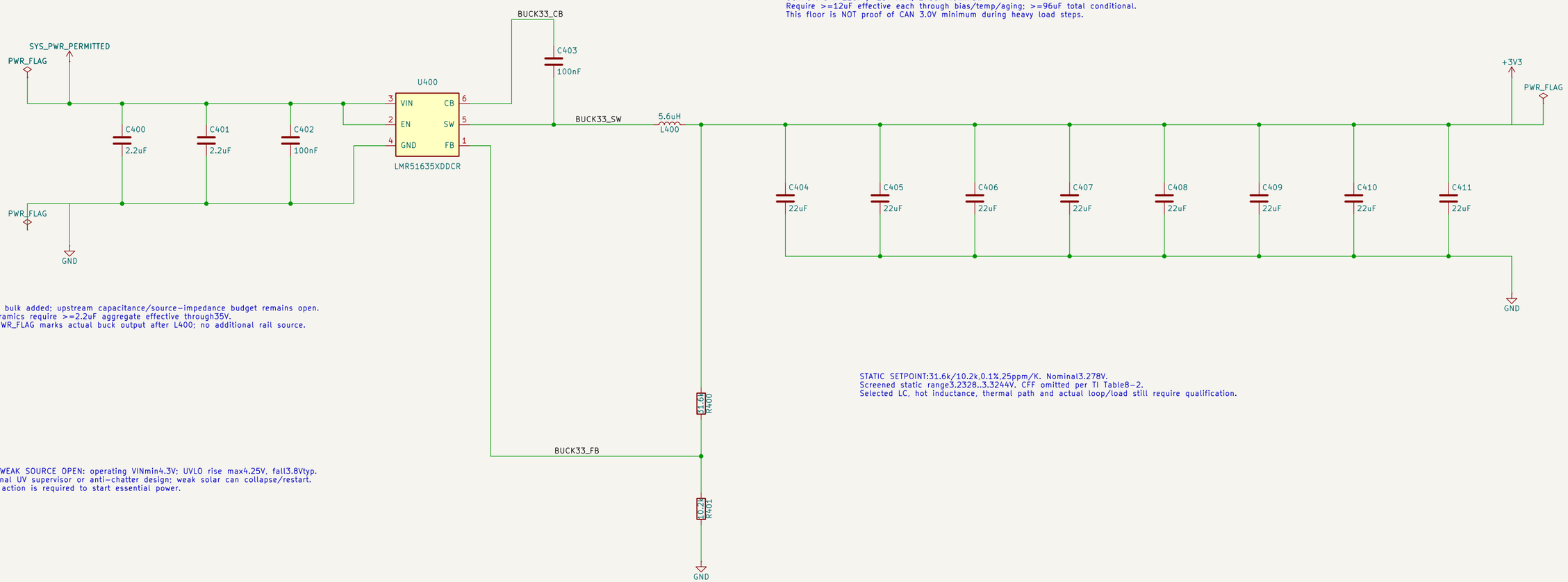
Existing TCAN custom symbols and manufacturer patterns reused.
Pin maps verified; assembly, ESD, timing and physical placement not qualified.
S08 construction review / S11 field-display limitations remain open.

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ESSENTIAL 3V3 — LMR51635XDDCR INTEGRATED DEVELOPMENT

EXTERNAL FIXTURE: SYS_PWR_PERMITTED and GND represent a bounded permitted source.
Input/return PWR_FLAGs do not implement source isolation, RBF, charger or battery protection.

5..35 V development envelope; 4.5 V sensitivity. EN = VIN; starts without MCU.
2 A is steady sizing only. Actual MCU/IHU/CAN ramp/load-step budget and transient acceptance remain open.



No input bulk added; upstream capacitance/source-impedance budget remains open.
Input ceramics require >=2.2uF aggregate effective through35V.
Output PWR_FLAG marks actual buck output after L400; no additional rail source.

OUTPUT: 8 x 22uF / 25V X7R, 176uF nominal.
Require >=12uF effective each through bias/temp/aging; >=96uF total conditional.
This floor is NOT proof of CAN 3.0V minimum during heavy load steps.

STATIC SETPOINT:31.6k/10.2k,0.1%,25ppm/K. Nominal3.278V.
Screened static range3.2328..3.3244V. CFF omitted per TI Table8-2.
Selected LC, hot inductance, thermal path and actual loop/load still require qualification.

UVLO / WEAK SOURCE OPEN: operating VINmin4.3V; UVLO rise max4.25V, fall3.8Vtyp.
No external UV supervisor or anti-chatter design; weak solar can collapse/restart.
No MCU action is required to start essential power.

OPEN AUTONOMOUS 3V3 OV CONTAINMENT: failed-short high-side can expose MCU/CAN
and25V output bank to-35V. Upstream solar OV protection does not contain this fault.
This stage does not establish complete EPS essential-power acceptance.

POLICY OPEN: S03 native severity and S11 field-display APIs unavailable.
Existing custom U400 reused; S08 original review-marker coverage unresolved. No PCB synchronization.

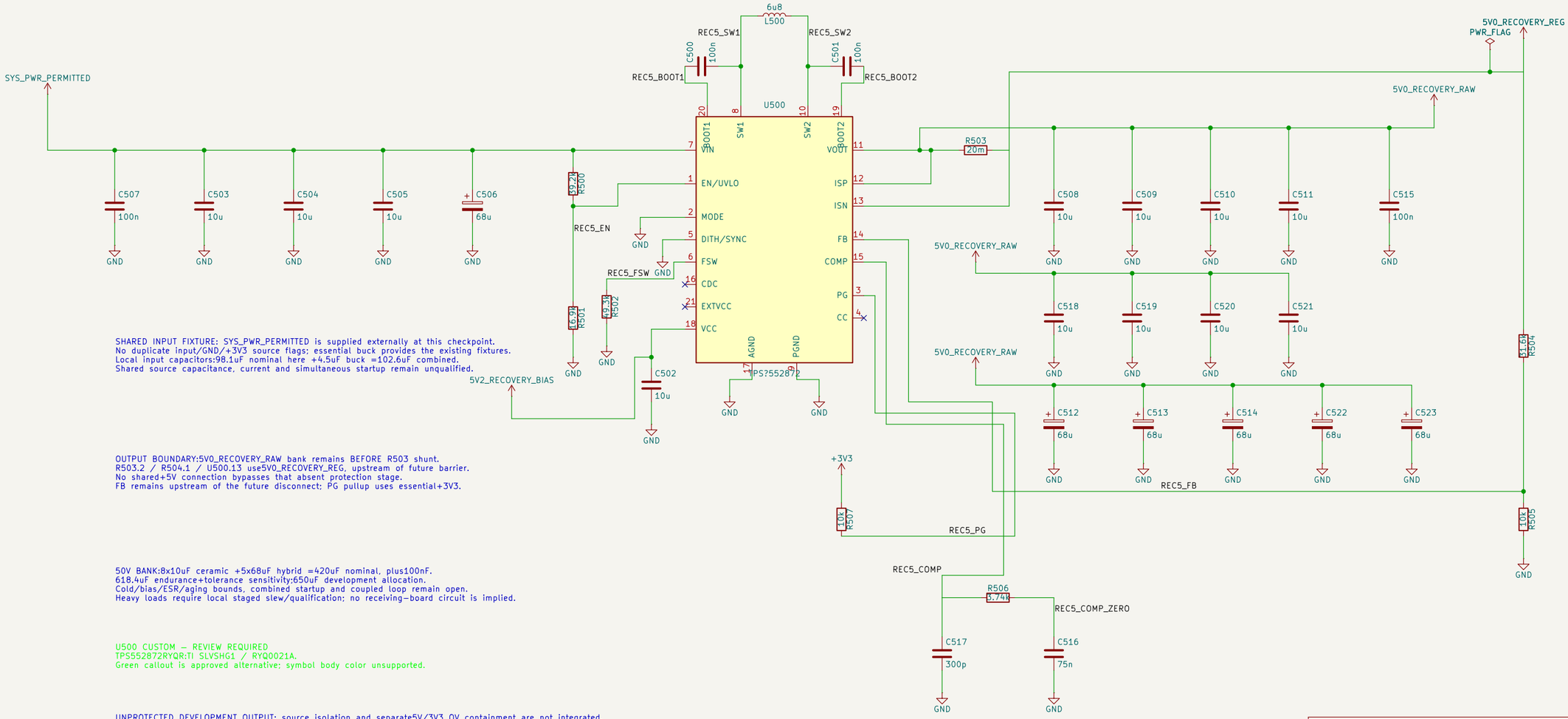
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RECOVERY 5V CONVERTER — INTEGRATED DEVELOPMENT

TPS552872RYQR: autonomous EN/UVLO, internal bias via floating EXTVCC; MODE low (PFM).
4.5..35 V development envelope: 2 A steady target only. Source/thermal/dynamic qualification remains open.



SHARED INPUT FIXTURE: SYS_PWR_PERMITTED is supplied externally at this checkpoint.
No duplicate input/GND/+3V3 source flags; essential buck provides the existing fixtures.
Local input capacitors:98.1uF nominal here +4.5uF buck =102.6uF combined.
Shared source capacitance, current and simultaneous startup remain unqualified.

OUTPUT BOUNDARY:5V0_RECOVERY_RAW bank remains BEFORE R503 shunt.
R503.2 / R504.1 / U500.13 use5V0_RECOVERY_REG, upstream of future barrier.
No shared+5V connection bypasses that absent protection stage.
FB remains upstream of the future disconnect; PG pullup uses essential+3V3.

50V BANK:8x10uF ceramic +5x68uF hybrid =420uF nominal, plus100nF.
618.4uF endurance+tolerance sensitivity:650uF development allocation.
Cold/bias/ESR/aging bounds, combined startup and coupled loop remain open.
Heavy loads require local staged slew/qualification; no receiving-board circuit is implied.

U500 CUSTOM — REVIEW REQUIRED
TPS552872RYQR:TI SLVSHG1 / RYQ0021A.
Green callout is approved alternative; symbol body color unsupported.

UNPROTECTED DEVELOPMENT OUTPUT: source isolation and separate5V/3V3 OV containment are not integrated.
S03 native severity and S11 field presentation APIs remain unavailable. No PCB synchronization.

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